

Introduction to Statistical Methods - Mid-Semester 1 Shorts

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Session 1 - Descriptive Statistics and Probability Framing

[Know More Detailed External Sources](#)

Intuition

First identify whose values matter, then describe center, spread, and shape together.

Definitions and jargon

A population is the full group; a sample is the observed subset. Population mean/SD/variance are μ, σ, σ^2 ; sample mean/variance/SD are $\bar{x}, s^2, s$. The five-number summary is minimum, lower quartile Q_1 , median, upper quartile Q_3 , maximum. Σ sums observations x_i ; i indexes them; n is their count; and interquartile range (IQR) is $Q_3 - Q_1$. Also, P means probability, X random variable, $||$ absolute distance, k number of SDs, and ∞ no finite upper limit.

Formulas, symbols, shapes, and assumptions

Session 1 compact formulas

$$\bar{x} = \frac{\sum x_i}{n} \quad s^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1}$$

$$\text{IQR} = Q_3 - Q_1 \text{ fences: } Q_1 - 1.5 \text{ IQR}, Q_3 + 1.5 \text{ IQR}$$

$$P(|X - \mu| < k\sigma) \geq 1 - \frac{1}{k^2} \text{ for } k > 1 \text{ and } 0 < \sigma^2 < \infty$$

Use median-of-halves excluding the overall median. The 68-95-99.7 rule needs approximate normality; Chebyshev is a guaranteed lower proportion for finite positive variance and $k > 1$.

Conditional properties

Weights need nonnegative values and positive total. Sample variance uses $n-1$. Pearson skew $3(\bar{x} - \text{median})/s$ is undefined at $s=0$; zero does not prove symmetry.

Important distinctions

Values beyond the IQR fences are flagged; whiskers usually end at observed values inside them. A high outlier can pull the mean more than the median.

Checked example

For 3, 5, 7, 9, 16, median 7, half-medians 4, 12.5, and IQR 8.5 give fences -8.75, 25.25; 16 is not flagged.

Traps and counterexamples

Do not apply the empirical rule to arbitrary skew or equate every maximum with a whisker.

Answer-free recall

Without notes: when prefer median/IQR; state both variance denominators; state exactly what Chebyshev guarantees.

Session 2 - Probability Axioms, Events, and Counting

[Know More Detailed External Sources](#)

Intuition

A probability model assigns consistent numbers from 0 to 1. Counting supplies a denominator only for equally likely outcomes.

Definitions and jargon

The sample space S contains all outcomes. Events A, B, A_i are subsets; i is an index; an elementary event has one outcome. $\cup$ / $\cap$ mean union, $\cap$ intersection, $\emptyset$ empty event, and Σ sum. $P(A)$ is event probability. For counting, n is available, r selected, $!$ factorial, $P(n, r)$ ordered selections, and $C(n, r)$ unordered selections.

Formulas, symbols, shapes, and assumptions

Session 2: probability and counting

$$P(A) \geq 0 \quad P(S) = 1 \quad \text{disjoint } A_i: P(\cup A_i) = \Sigma P(A_i)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(n, r) = \frac{n!}{(n-r)!} \quad C(n, r) = \frac{n!}{r!(n-r)!}$$

Multiply successive choice counts. Use permutations for order and combinations for membership, with integers $0 \leq r \leq n$.

Conditional properties

Independence means $P(A \cap B) = P(A)P(B)$; exclusivity means $P(A \cap B) = 0$. Thus positive-probability exclusive events are dependent.

Important distinctions

Braces list outcomes, so $\{2, 4, 6\}$ is composite. Replacement can preserve draw probabilities; order matters for roles, not committees.

Checked example

Choosing 3 of 9 applicants gives $C(9, 3) = 84$; assigning chair, secretary, treasurer gives $P(9, 3) = 504$.

Traps and counterexamples

The empty event is $\emptyset$, not a one-element set. Subtract overlap in a union. A finite sample space need not be equally likely.

Answer-free recall

Without notes: state the axioms; contrast independence/exclusivity; give a permutation-not-combination story.

Session 3 - Conditional, Joint, and Total Probability

[Know More Detailed External Sources](#)

Intuition

Conditioning shrinks the reference group. A table column, a tree branch, and the event after the vertical bar all identify the new denominator.

Definitions and jargon

A joint probability concerns events together; a marginal totals other categories; a conditional restricts attention; a partition is nonoverlapping and exhaustive. $P(E)$ means probability of event E ; A, B, C are events; $|$ means “given”; $\cap$ intersection; B_i partition branch i ; and Σ sum over branches.

Formulas, symbols, shapes, and assumptions

Session 3: condition, multiply, then add paths

$$P(A | B) = \frac{P(A \cap B)}{P(B)} \quad P(B) > 0$$

$$P(A \cap B \cap C) = P(A)P(B | A)P(C | A \cap B)$$

$$\text{Partition } B_i: P(A) = \Sigma P(B_i)P(A | B_i)$$

Multiply along a branch for a joint; add disjoint branches for a marginal; divide joint by conditioning marginal to reverse direction.

Conditional properties

$P(A|B)$ needs $P(B)>0$. Total probability needs exclusive, exhaustive branches; each chain factor conditions on all earlier branch events.

Important distinctions

$P(A|B)$ usually differs from $P(B|A)$. A marginal uses the full total; a conditional uses the fixed row or column total.

Checked example

Of 120 commuters, 72 use buses and 18 of those are late. Of the 48 others, 6 are late. Thus $P(\text{late})=24/120=0.20$, while $P(\text{bus} | \text{late})=18/24=0.75$. The first denominator is everyone; the second is only late commuters.

Traps and counterexamples

Overlapping paths double-count; multiplying marginals needs independence; conditioning on probability zero is undefined.

Answer-free recall

Without notes: explain the denominator; state the three-event chain rule; list both partition requirements.

Session 4 - Bayes Theorem and Naive Bayes

[Know More Detailed External Sources](#)

Intuition

Bayes reverses a condition: weight each prior cause by its likelihood, then normalize across all causes.

Definitions and jargon

$P(E)$ is probability of event E ; C is class, x evidence, $|$ “given,” and $\prod$ indexed multiplication. Prior $P(C)$, likelihood $P(x|C)$, evidence $P(x)$, and posterior $P(C|x)$ are Bayes's four quantities.

Formulas, symbols, shapes, and assumptions

Diagram: A is evidence; B_i partition class i ; B_j target class j ; k class count; Σ sum. Classifier: C_j is class j , x_i feature i , p feature count.

$$P(B_j | A) = \frac{P(B_j)P(A | B_j)}{\sum_{i=1}^k P(B_i)P(A | B_i)}$$

Naive Bayes score is $P(C_j)\prod_i P(x_i|C_j)$. Compare scores to classify; normalize for posteriors. Logs replace products with sums and reduce numerical underflow.

Conditional properties

Assume $P(x_1, \dots, x_p | C) = \prod_i P(x_i | C)$, not unconditional independence. Additive smoothing prevents unseen-category zeros; adjust denominators.

Important distinctions

Prior affects posterior. Argmax is the largest-score class. Omit shared evidence only for argmax, not normalized probabilities.

Checked example

Two classes have priors 0.6 and 0.4. Evidence likelihoods are 0.2 and 0.5. Scores are 0.12 and 0.20; evidence is 0.32, so the second posterior is $0.20/0.32=0.625$.

Traps and counterexamples

Do not reverse likelihood and posterior. One zero erases an unsmoothed product; correlated features can violate the factorization.

Answer-free recall

Without notes: name the four Bayes quantities; locate the independence assumption; say when evidence may be omitted.

Session 5 - Random Variables and Joint Distributions

[Know More Detailed External Sources](#)

Intuition

A random variable maps outcomes to numbers; its distribution assigns point mass or continuous density.

Definitions and jargon

A probability mass function (PMF) assigns discrete mass; a probability density function (PDF) continuous density; a cumulative distribution function (CDF) probability through a value. X, Y are random variables and x, y their values. P means probability; p mass/joint mass; f density; $F(x) = P(X \leq x)$ the CDF; $|$ "given." Σ means sum, $\int$ integral, dx integration over x , E expectation, Var variance; subscripts name variables.

Formulas, symbols, shapes, and assumptions

Session 5: mass, density, moments, and joints

PMF: $p(x) \geq 0, \Sigma p(x) = 1$ PDF: $f(x) \geq 0, \int f(x) dx = 1$

$E[X] = \Sigma x p(x)$ or $\int x f(x) dx$ $\text{Var}(X) = E[X^2] - E[X]^2$

$p_x(x) = \Sigma_y p_{x,y}(x, y)$ $p(x | y) =$

$$\frac{p(x, y)}{p_y(y)}$$

Independence requires joint factorization; zero covariance alone does not.

Moments require finite sums/integrals. For continuous X , $P(X=x)=0$; interval probability is area.

Conditional properties

Joints are nonnegative and normalize to 1. Marginalize out a variable; independence requires factorization on all support.

Important distinctions

Mass differs from density height. $E[X^2]$ need not equal $E[X]^2$. Independence implies zero covariance, not conversely.

Checked example

Masses .2, .5, .3 at 0, 1, 2 give $E[X]=1.1$, $E[X^2]=1.7$, and $\text{Var}(X)=.49$.

Traps and counterexamples

A CDF cannot decrease; marginals alone do not determine dependence.

Answer-free recall

Without notes: state both normalization rules; recover a marginal; relate independence and zero covariance exactly.

Session 6 - Named Probability Distributions

External Sources

Intuition

Choose from the data story: one binary trial, successes in fixed trials, events per exposure, or a continuous measurement.

Definitions and jargon

X is the random variable; x, k possible values; n trial count; p success probability; P probability; E expectation; Var variance. $C(n, k)$ chooses k from n ; $!$ is factorial: $n! = n(n-1) \cdots 1$, $0! = 1$. λ (lambda) is Poisson expected count; e exponential constant. Normal mean/standard deviation (SD) are μ, σ with $\sigma > 0$; Z is standardized. $\in$ means “belongs”; $\sim$ means “has distribution.” Degrees of freedom count independent information after estimation.

Formulas, symbols, shapes, and assumptions

Session 6: match the data-generating story

Bernoulli: $x \in \{0, 1\}$, $E[X] = p$, $\text{Var}(X) = p(1-p)$

Binomial: $P(X=k) = C(n, k)p^k(1-p)^{n-k}$, $k=0, \dots, n$

Poisson: $P(X=k) = e^{-\lambda} \frac{\lambda^k}{k!}$ $k=0, 1, \dots$; $E[X] = \text{Var}(X) = \lambda$

Normal: $Z = \frac{X-\mu}{\sigma}$ support all real, $\sigma > 0$

t: all real; chi-square and F: nonnegative.

Bernoulli is one binary trial; binomial counts fixed independent equal- p trials; Poisson counts a fixed exposure; normal is continuous.

Conditional properties

Binomial needs fixed n , independence, two outcomes, constant p ; exposure rescales Poisson λ . Small-sample t, chi-square, and F require their stated normal-sample/independence constructions.

Important distinctions

Poisson support is unbounded; binomial stops at n . Normal/t support all reals; chi-square/F are nonnegative and usually right-skewed.

Checked example

For $X \sim \text{Binomial}(6, 0.3)$, $P(X=1) = C(6, 1)(0.3)(0.7^5) \approx 0.3025$, $E[X] = 1.8$, and $\text{Var}(X) = 1.26$.

Traps and counterexamples

Binary data do not prove independent constant- p trials; equal moments do not prove Poisson. No hypothesis testing here.

Answer-free recall

Recall all seven supports/parameters, binomial conditions, and Poisson exposure scaling.

Session 7 - Sampling, the CLT, and Confidence Intervals

[External Sources](#)

Intuition

A sampling distribution tracks a statistic over samples; a confidence interval (CI) combines estimate with uncertainty.

Definitions and jargon

n is sample size. A simple random sample (SRS) gives each size- n subset equal chance; stratified sampling samples within every stratum/group. SD means standard deviation, SE standard error, iid independent and identically distributed, CLT central limit theorem. $\bar{X}$ is sample mean; μ population mean; σ population SD; s sample SD; E expectation; Var variance; σ^2 population variance; α noncoverage. The diagram's z subscript $1-\alpha/2$ selects the standard-normal upper quantile; its t subscripts $1-\alpha/2, n-1$ select the Student's t upper quantile with $n-1$ degrees of freedom. These are critical values z^*, t^* . $N(0,1)$ is standard normal; $\sqrt{}$ square root; $\pm$ plus/minus. The $E[\bar{X}]$, $\text{Var}(\bar{X})$, SE, and CLT formulas require iid observations with finite variance; dependence may require adjustment.

Formulas, symbols, shapes, and assumptions

Session 7: repeated-sample behavior

$$E[\bar{X}] = \mu \quad \text{Var}(\bar{X}) = \frac{\sigma^2}{n} \quad SE(\bar{X}) = \frac{\sigma}{\sqrt{n}}$$

$$\text{CLT: } \frac{\bar{X} - \mu}{\sigma / \sqrt{n}} \text{ approaches } N(0,1)$$

$$\text{Known } \sigma: \bar{X} \pm z_{1-\alpha/2} \frac{\sigma}{\sqrt{n}} \quad \text{Unknown } \sigma: \bar{X} \pm t_{1-\alpha/2, n-1} \frac{s}{\sqrt{n}}$$

Coverage belongs to the method; a realized frequentist interval assigns no probability to fixed μ .

Conditional properties

Known σ : z^* ; unknown: Student's t^*, s . Small-sample exactness needs normal iid data.

Important distinctions

Population/sample/sampling distributions differ. SD describes individuals; SE estimators. No universal CLT $n=30$.

Checked example

For independent $n=49$, $\bar{X}=18$ ms, known $\sigma=3.5$ ms, $z^*=1.96$: $SE=.5$ ms; 95% CI (17.02, 18.98) ms.

Traps and counterexamples

A realized CI gives no probability about fixed μ ; convenience bias remains.

Answer-free recall

Recall SRS/stratified, mean moments/SE, and frequentist CI meaning.

Session 8 - Integrated Review and Method Selection

External Sources

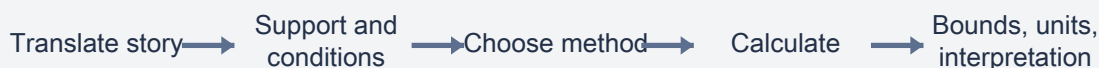
Intuition

Most errors begin with the wrong denominator, support, target, or assumption. Translate before calculating.

Definitions and jargon

The target is what is asked; support lists possible values. A validity condition enables a rule; a sanity check tests bounds or units. SD means standard deviation and CI confidence interval. For a Poisson count, λ (lambda) is expected count for its exposure; $|$ means “given.”

Formulas, symbols, shapes, and assumptions



Name variables first. Means/SDs retain units, variances squared units, probabilities no units. State independence or partition structure.

Conditional properties

The empirical rule needs approximate normality; Chebyshev finite positive variance. Total probability needs a partition; binomial every trial condition; a mean CI valid design/model conditions.

Important distinctions

Description summarizes; inference generalizes under a sampling model. Counts need justified equally likely outcomes. Fence is not whisker; likelihood not posterior.

Checked example

A Poisson process averages 4 calls per 30 minutes. In 90 minutes $\lambda=12$; mean and variance are 12, with support $0, 1, 2, \dots$.

Traps and counterexamples

Reject negative variance, probability outside $[0, 1]$, count outside support, or a CI treated as certainty. Recheck the condition after $|$ and the unit scale.

Answer-free recall

Recall the workflow, one decisive condition per session, and three sanity checks.